

Kevin Jaewoo Lee

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PROJECT

Ticket Booking Web Service

- Developed a highly consistent ticket-booking backend using DynamoDB with Node.js in a serverless environment with Lambda and API Gateway, implementing TTL-based seat locks to handle real-time reservations reliably.
- Optimized search engine by indexing eventName and location queries using Amazon OpenSearch. Designed query and keyword custom analyzers(edge-ngram, autocomplete, synonyms) to detect full-text and structured filtering for ticket-booking data.
- Implemented presigned URLs to securely upload event posters and seat maps directly from clients to Amazon S3, reducing average TTFB from 600.7 ms to 30.2 ms, achieving a 94.5% improvement.

University Course Selection Application

- Developed a full-stack web application using Express.js and React.js, integrating Firebase for user registration and authentication.
- Integrated with MongoDB, enabling user data, course selection, and chatroom service to interact and manage, allowing users to register for courses and communicate within the corresponding course.
- Designed a real-time chat service using WebSockets to support instant message broadcasting and tracking user presence, enabling seamless communication between users enrolled in the same course.

Notice Board Web Application

- Developed a Notice Board application using Spring Boot, Spring Data JPA, and H2 Database, implementing complete CRUD operations with the MVC pattern.
- Integrated Thymeleaf as a view template engine to dynamically render HTML pages, using DTOs and Entities for secure data transfer.
- Implemented composite indexes on frequently queried paths (e.g., author_id, post_id, created_at) to improve query performance, reducing P95 read latency by 35%.

WORK EXPERIENCE

Software Developer

03/2025 - 08/2025

Grow Your Business

Norwalk, Connecticut

- Developed a real-time chat service powered by Firebase Firestore and React, implementing thread creation and snapshot listeners for seamless updates. Refactored chat front-end to state-driven rendering model, restructuring the component to ensure stable and smooth multi-user interactions.
- Refactored a single-assistant model into multiple role-specialized agents using OpenAI API with custom fine-tuning to reflect company tone, goal, and product knowledge, improving response accuracy and efficiency .
- Implemented secure authentication and authorization using Spring Security, and JWT to replace client-side login logic, enabling OAuth 2.0 Google Login and token-based session management; Configured global CORS policy and custom JWT validation filters, ensuring secure communication between backend and frontend.

TECHNICAL SKILLS

Languages: C++, C, JavaScript, TypeScript, Python, React.js, HTML, CSS

Technologies: MySQL, Node.js, WebSocket, Firebase, JWT, Spring Boot, Spring Security, OpenAI API, Fine-tuning

AWS: Lambda, DynamoDB, S3, Cloudfront, OpenSearch, API Gateway

EDUCATION

University of California San Diego

B.S in Computer Science & Software Engineering

09/2024 - 05/2026

GPA: 3.90

Diablo Valley College

A.S in Computer Science

08/2022 - 05/2024

GPA: 3.95